MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, COUNTERS,
MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/13F, dated 16 April 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.1, delete and substitute:

"1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, binary and decode counters. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number."

* 1.2.3, delete letter B in its entirety, and substitute "F-3 (14-lead, 3/16" x 1/4"), flat package".

PAGE 2

* 1.3, delete "Thermal resistance, junction to case (θ_{JC}) 0.09° C/mW for flat package; 0.08°C/mW for dual-in-line package" and substitute:

"Thermal resistance, junction to case (θ_{JC})- - - - - - (See MIL-M-38510 Cases A, B, C, D, E, F- - - - - - - - - - - - - - - - - - Appendix C)."

* 1.3, after Junction temperature (T_J): Add "2/".

* 1.4, after Supply voltage: Add "(V_{CC})".

* 1.4, Width of clock pulse, $t_{w(CLOCK)}$ device types 03, 04, 05, 06: Delete "30 ns" and substitute "25 ns".

1.4, setup time, device types 03, 04, 05, 06: Delete "Enable P - - - - - 25 ns minimum".

1.4, following hold time operating conditions, add enable time conditions as follows: "Enable time, t(Enable), device types 03, 04, 05, 06, Enable P ------ 25 ns minimum".

* 1.4, following footnote 1/, add:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening condition per Method 5004 of MIL-STD-883."

PAGE 3

* 3.2.3, delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request."

* 3.6, delete and substitute:

"3.6 Marking. Marking shall be in accordance with MIL-M-38510."

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PAGE 4

TABLE I, I_{IL5}, device type 08, 09, Min limits column: Delete "-0.7" and substitute "-0.5".

PAGE 7

* 4.2.c, delete and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

PAGE 8

* 6.1: Delete in its entirety.

* 6.4: GND - - - - - Delete in its entirety and substitute "Ground zero voltage potential."

PAGES 38 AND 39

FIGURES 6 and 7, Note 1, delete "PRR = 1 MHz" and substitute "PRR $\leq$ 1 MHz".

PAGE 41

* FIGURE 8A, delete and substitute new figure 8A as printed on page 5 of this amendment.

PAGE 42

* FIGURE 8B, delete and substitute new figure 8B as printed on page 6 of this amendment.

PAGE 43

* FIGURE 8C, Note 1, delete "t_p = 20 ns" and substitute "t_p = 25 ns".

FIGURE 8C, Note 1, delete "V_{GEN} = 4.5 V" and substitute "V_{GEN} = 3.0 V".

PAGES 44, 45, AND 46

FIGURES 9, 10, and 10A, Note 1, delete "PRR = 1 MHz" and substitute "PRR $\leq$ 1 MHz".

PAGE 49

TABLE III, footnote 14/, delete and substitute as follows:

"14/ The limits shown shall be as follows:

Test	Min/Max limits (mA) for circuit:				
	A	B	C	D	E
I_{IL1}	-0.4/-1.3	-0.4/-1.3	-0.3/-1.3	-0.7/-1.6	-0.7/-1.6
I_{IL2}	-1.4/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2
I_{IL3}	-1.4/-4.8	-0.7/-4.8	-0.7/-4.8	-1.4/-6.4	-1.4/-6.4

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PAGE 52

TABLE III, footnote 14/, delete and substitute as follows:

"14/ The limits shown shall be as follows:

Test	Min/Max limits (mA) for circuit:				
	A	B	C	D	E
I_{IL1}	-0.4/-1.3	-0.4/-1.3	-0.4/-1.3	-0.7/-1.6	-0.7/-1.6
I_{IL2}	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2
I_{IL3}	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2

"

PAGE 53

TABLE III, tests 43 and 45, Min test limits column: Add "9/ after "-0.7". Make this same change on pages 58 and 68.

PAGE 57

TABLE III, footnotes: Add new footnote 9/ as follows:

"9/ Min limits for circuit C shall be -0.5." Make this same change on pages 62, 67, and 72.

PAGE 63

TABLE III, tests 43 through 45, Min test limits column: Add "9/ after "-0.7".

PAGE 72

TABLE III, subgroup column: Add "11"; conditions: Add "Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C".

PAGES 80 AND 86

TABLE III, Device types 08 and 09, Subgroup column: Add "8"; Conditions: Add "Repeat subgroup 7 at T_C = 125°C and T_C = -55°C".

PAGE 82

TABLE III, test 6, symbol column: Delete "I_{IL8}".

PAGES 85 AND 86

TABLE III, Subgroup column: Delete "T_C = 125°C" and substitute "T_C = 25°C".

PAGE 87

TABLE III, footnote 4/, delete and substitute "See figure 10A. Apply waveform E for tests 161 and 178, and waveform H for tests 162 and 179."

PAGE 88

TABLE IV, Circuit B column: Add manufacturers designation for device type 09.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS, TD
Air Force - 11, 19, 85, 99
DLA - ES

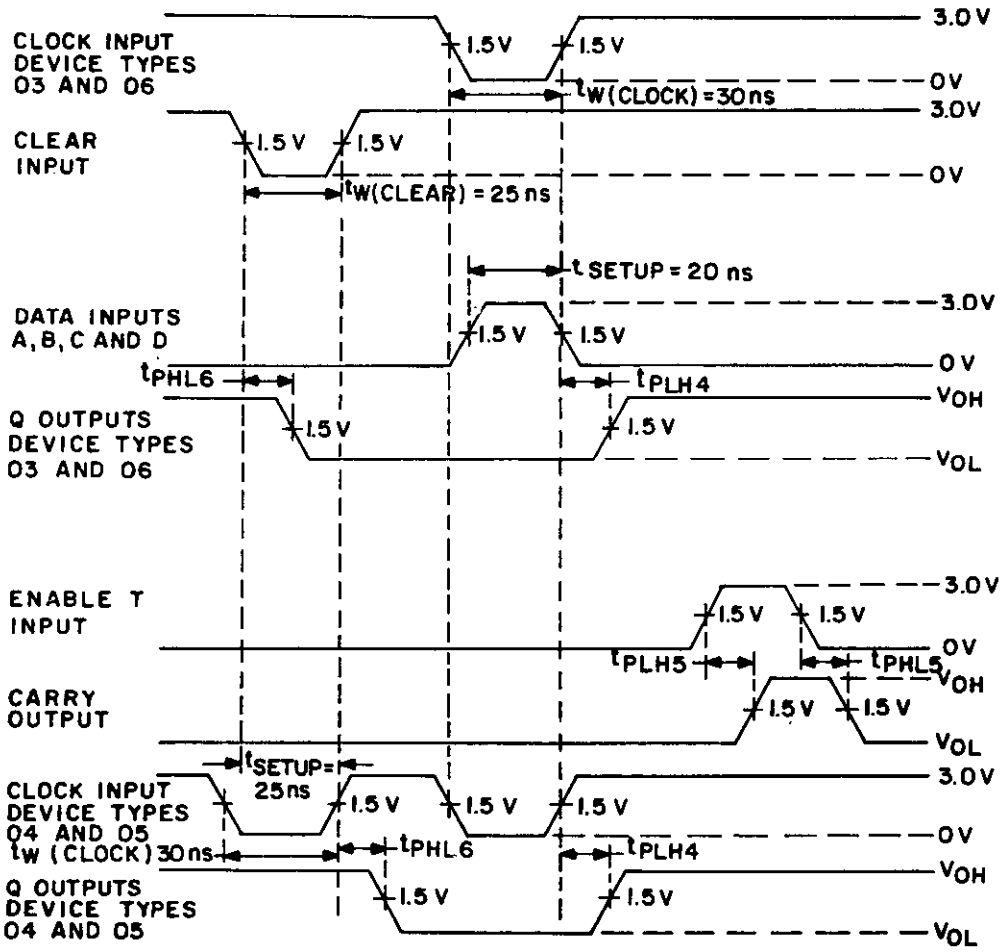
User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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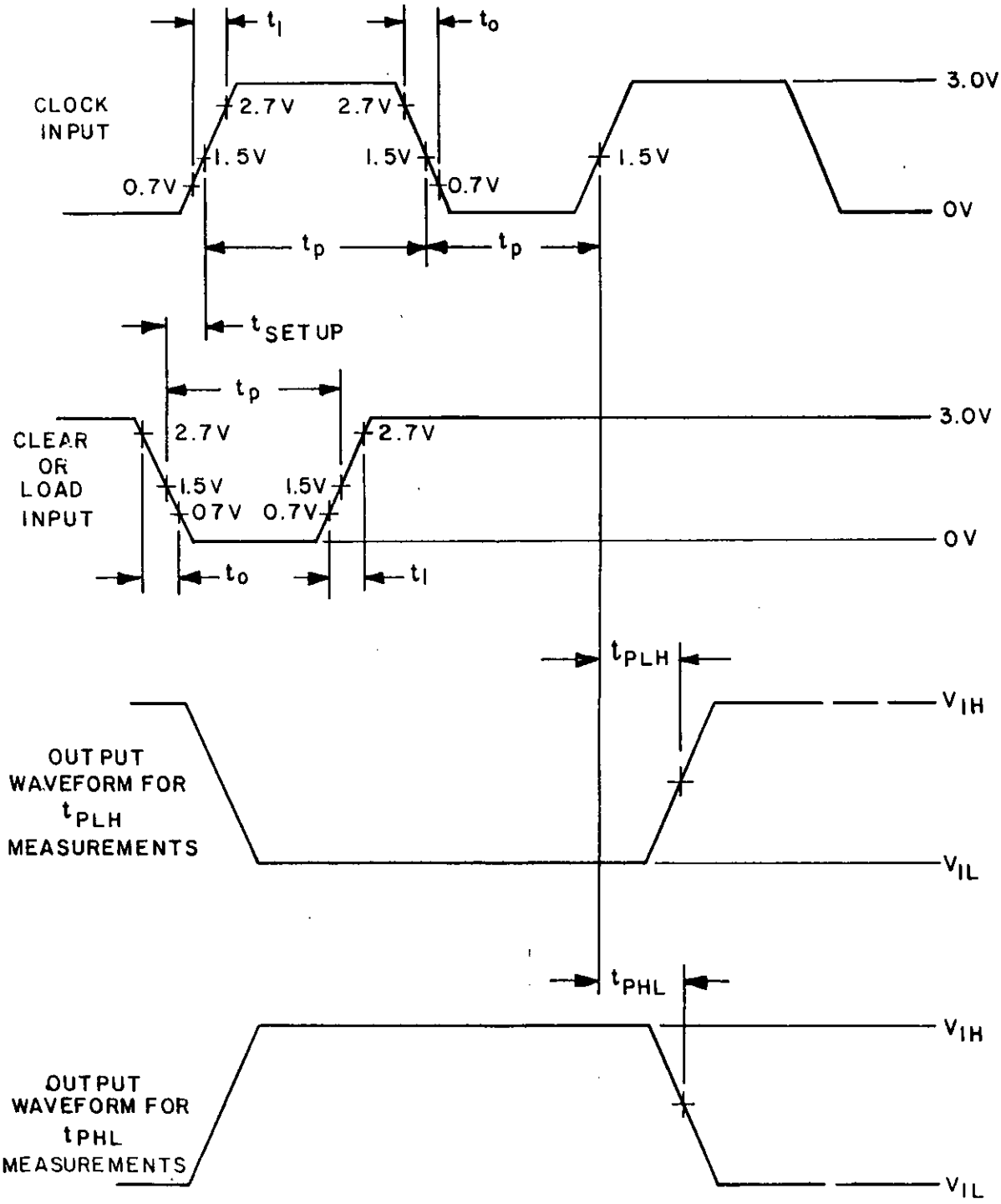


VOLTAGE WAVEFORMS

NOTES:
1. The input pulses are supplied by a generator having the following characteristics: $t_r \leq 10$ ns, $t_f \leq 10$ ns, PRR $\leq$ 1 MHz, Z_{OUT} 50Ω.
2. Enable T duty cycle ±50%, PRR 1 MHz.

FIGURE 8A. Switching time test circuits and waveforms for device types 03, 04, 05, and 06 - Continued.

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NOTES:
1. The input pulses are supplied by generators having the following characteristics: $t_0 = t_1 \leq 10$ ns, PRR ≤ 1.0 MHz, $t_p \geq 500$ ns, $Z_{OUT} \approx 50\Omega$.
2. The t_{SETUP} for load pulse = 30 ns; t_{SETUP} for clear pulse = 25 ns.

FIGURE 8B. Switching time test circuits and waveforms for device types 03, 04, 05, and 06.

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